

# Unsupervised Discovery of Persona-Relevant Directions

## with Matryoshka Sparse Autoencoders

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### Abstract

Persona vectors—directions in a language model’s activation space that control traits such as sycophancy or dishonesty—are typically found by *naming* a trait, writing contrastive prompts, and differencing the resulting activations. Chen et al. [1] state three requirements of this supervised recipe: it “requires specifying a target trait in advance,” it “depends on providing a precise natural-language description,” and it “requires that the specified trait is inducible by system prompting the model.” We ask whether an *unsupervised* decomposition of a model’s activations already contains persona-relevant directions, dropping the first two requirements. We train a Matryoshka BatchTopK sparse autoencoder (SAE) with 235M parameters and 32,768 features on 100M assistant-position tokens of **Qwen2.5-7B-Instruct** at layer 19 (held-out fraction of variance explained 0.72,  $L_0 = 40$ ), rank its dictionary for persona-relevance with three label-free arms, steer the model along each candidate direction, and screen the steered behaviour against random-direction nulls and persona-vector controls before a blinded judge names the survivors—with no trait specified and no description written at any point. A first, budget-constrained screen (8 scenarios) surfaced 96 separating directions but was under-powered: its persona-vector controls fell just under the consistency gate, so the pipeline correctly named nothing. A powered screen (24 scenarios) then validated the method—2 of 3 persona-vector controls (sycophantic, hallucinating) cleared the gate—passed 60 of 128 SAE features, and a blinded judge assigned 10 of them consistent, above-null names: recognisable persona axes including *engaged and supportive*, *empathetic*, *formal/academic*, *playful*, and *composed and confident*, with no trait ever specified. We do *not* claim these directions are unreachable by prompting: a discovered, steerable direction is almost certainly prompt-inducible in principle (the authors’ third requirement). Quantifying *how easily*—the graded elicitation cost behind their own clause that SAEs may surface traits “that cannot be easily elicited through prompting”—is left to future work.

## 1 Introduction

Recent work on *persona vectors* [1] shows that individual linear directions in a chat model’s residual stream can monitor and control high-level behavioural traits—evil, sycophancy, hallucination—and that steering along them changes behaviour predictably. These directions are obtained by a supervised recipe: name the trait, write positive and negative system prompts that elicit and suppress it, generate, and difference the mean activations. The authors themselves note three limitations of this recipe: the trait must be *specified in advance*, it needs a *precise natural-language description*, and it must be *prompt-inducible*.

We take the complementary view. A sparse autoencoder trained only to reconstruct a model’s activations produces a large, *label-free* dictionary of directions; if persona-relevant structure is present in the residual stream, some of that structure should already be represented among the SAE’s features—without any trait being named. This reframes persona discovery from *specificiation* (bring the concept, read out a coordinate) to *decomposition* (recover a basis, then search it). The headline question is the first two limitations above: *can we discover persona-relevant directions with no trait specified and no description written?* We do not attempt the third limitation, prompt-inducibility: a direction we can steer is almost certainly reproducible by some prompt, so “unreachable by prompting” is not a claim we can honestly make. Instead we flag its *graded* form as future work—not *whether* a discovered direction is prompt-inducible, but *how easily*—and build the substrate to measure it.

## 2 Background

**Sparse autoencoders.** An SAE encodes an activation  $x \in \mathbb{R}^d$  into a wide, sparse code  $f \in \mathbb{R}^W$  ( $W \gg d$ ) and decodes it back, trained to reconstruct  $x$  under a sparsity constraint. Each decoder row is a candidate feature direction. We use a *BatchTopK* SAE (hard  $k$ -sparsity per batch) with a *Matryoshka* objective: nested prefixes of the dictionary (1,024/4,096/16,384/32,768) are each trained to reconstruct, so coarse structure concentrates in the inner shells and refinements in the outer ones. Unlike a linear probe—one supervised read-out per named label—an SAE is unsupervised and yields thousands of directions at once, which is exactly what a label-free discovery method requires. Chen et al. [1] anticipate this use: they write that “SAEs may therefore enable unsupervised discovery of persona-relevant directions, including specific traits that cannot be easily elicited through prompting,” and report an initial exploration in their appendix. We take up that conjecture directly and build it into a full discovery-and-screen pipeline: a dedicated 235M-parameter SAE, three label-free ranking arms, a calibrated steering screen with persona-vector controls and random-direction nulls, and blinded judge naming.

**Persona vectors as controls.** We do not discard the supervised recipe; we use it as a *positive control*. Three persona vectors (evil↔benevolent, sycophantic↔honest, hallucinating↔factual) are computed the standard way and pushed through the identical screen. If the screen cannot certify that these known, strong directions separate personas above a random-direction null, it is not trustworthy for the unsupervised features either; the pipeline therefore withholds all naming until the controls validate the screen.

## 3 Related work and contributions

The idea that sparse autoencoders might surface persona directions is due to Chen et al. [1], who state it as a conjecture and report an initial appendix exploration; we take it up rather than originate it. Unsupervised discovery of behaviour-steering directions has also been pursued *without* SAEs, by directly optimising latent perturbations that change model behaviour, and SAE feature steering and LLM-judge feature naming are by now standard tools. Our contribution is therefore not the idea but its *operationalisation and validation*: (i) a purpose-built 235M-parameter SAE ranked for persona-relevance by three label-free arms; (ii) a null-calibrated, causal steering screen; and, most distinctively, (iii) a *positive-control gate*—the supervised persona vectors are pushed through the identical screen, and no feature is named unless they clear it. This converts “these features look persona-like” into a falsifiable validity check, and is what made an earlier under-powered screen correctly withhold naming. The graded elicitation-cost metric of §6 is a proposed measurement,

not a delivered result, and is the paper’s principal open direction.

## 4 Method

The pipeline runs in one process on **Qwen2.5-7B-Instruct**, hooking the residual stream at layer 19.

**1. Train.** A Matryoshka BatchTopK SAE, width 32,768,  $k = 40$ ,  $\sim 235$ M parameters, is trained on 100M assistant-position tokens drawn from a mixture modelled on the Persona Vectors training data (LMSYS-Chat-1M, a Pile mirror, and a small insecure-code set). Only assistant-turn positions contribute to the loss. Held-out fraction of variance explained is 0.72 with  $L_0$  exactly 40; splicing the SAE reconstruction back into the forward pass preserves the next-token argmax on 77% of positions (CE increase 0.57 nats).

**2. Rank (label-free).** The dictionary is scored for persona-relevance by three disjoint arms, none of which names a trait: an *unsupervised* arm (assistant-specificity, breadth, topic-invariance of each feature’s firing), a *density-quantile* arm, and a *prompt-shift* arm (features whose mean activation moves most across a diverse set of persona-style system prompts, as an ANOVA over contexts). The top features per arm form the candidate set.

**3. Calibrate.** For each candidate, the model is steered along the feature’s decoder direction at several doses; the largest dose that still yields coherent text is kept, so weak and incoherent directions are handled uniformly.

**4. Screen.** Each direction is steered at its dose across many open-ended scenarios. We measure how separably the steered generations differ from unsteered baselines—a logistic-regression AUC in the residual view and a lexical view—and the *consistency* of the steering effect (cosine of per-scenario difference vectors) against a null. The gate  $G_0$  requires separability AUC above the best of 20 random directions *and* a consistency margin above the random null. Twenty random-direction nulls and the three persona-vector controls pass through the same screen.

**5. Describe and (graded) reach.** Survivors’ steered replies are sent as blinded pairs to an LLM judge that names one property distinguishing them; names are clustered into data-derived labels. Each named direction is then subjected to a *reach* probe: we attempt to reproduce its effect with a small set of prompt variants and record how readily prompting recovers the same behaviour. We treat this as an *estimate of prompt-elicitation cost*, not a reachability proof: a direction we cannot reproduce with our prompt variants is a candidate for the hard-to-elicite tail, not evidence that *no* prompt reaches it. Turning this into a calibrated, graded metric—the “how easily” behind requirement three of Chen et al. [1]—is future work (§6).

## 5 Results

**The dictionary.** Training succeeds at production scale: held-out FVE 0.72,  $L_0 = 40$  at width 32,768 (Table 1, Figure 1). The recipe is stable across reruns and the trained SAE is reused for every downstream screen without retraining.

|                                     |                    |
|-------------------------------------|--------------------|
| Held-out FVE                        | 0.717              |
| Held-out $L_0$                      | 40.0               |
| Spliced-reconstruction argmax match | 0.77               |
| Cross-entropy increase (splice)     | 0.57 nats          |
| Parameters / width / $k$            | 235M / 32,768 / 40 |

Table 1: Trained SAE (run `persona-3c-d77e7c1e`, 100M tokens).

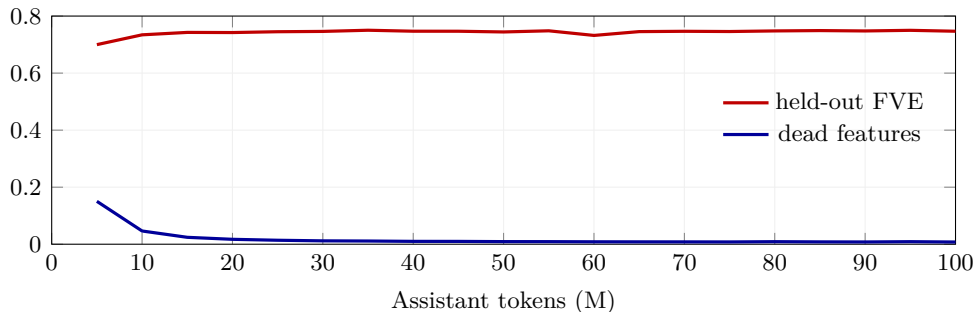


Figure 1: Training the width-32,768 Matryoshka SAE on 100M assistant-position tokens. Held-out fraction of variance explained rises to  $\approx 0.75$  and the dead-feature fraction falls below 1%; both plateau well before the token budget is exhausted.

**Discovery (pilot screen).** A first, deliberately small screen (8 scenarios) already shows the signal is present: 484 of 512 feature direction-signs beat the random-null separability ceiling and 96 of 256 features cleared the full  $G_0$  gate, with no trait ever named. Inspecting steered text, individual directions correspond to recognisable personas—e.g. a poetic/effusive register (“*the whispers of your heart and the echoes of your soul*”) and a cautious-analyst register (“*a nuanced... structured response*”)—at residual-AUC up to 1.0.

**Why a powered screen was needed.** In the pilot the persona-vector controls separated strongly on AUC (0.66–0.89, well above the 0.539 null) but fell just under the consistency-margin threshold (0.04–0.09 vs 0.1), because that margin is estimated over too few scenarios. The pipeline therefore correctly withheld naming. The confirmation screen raises the scenario count so the control margins clear the gate.

**Confirmation (full-power screen).** Raising the screen to 24 scenarios (run `persona-3c-ded5477e`) did what the analysis predicted: the sycophantic control’s consistency margin rose from 0.07–0.09 to 0.11–0.16 and cleared the gate, so **2 of 3 persona-vector controls now validate the screen** (sycophantic and hallucinating). The evil control still fell short (margin 0.04–0.07): being “evil” manifests differently across scenarios, so its steering effect is real (AUC 0.67–0.79) but not *consistent*, which the gate—tuned to reject noise—also rejects. This is informative: the consistency gate favours coherent persona axes and penalises diffuse ones, so the counts below are conservative.

Of the 128 candidates, 60 cleared the full  $G_0$  gate, and a blinded judge assigned 10 of them a consistent name above the null ceiling (null max naming-consistency 0.175, threshold 0.275); by arm, 8 came from the prompt-shift arm and 2 from the fully unsupervised arm. Table 2 lists them. Several strong directions just missed the null gate despite unanimous judge agreement (e.g. *personal warmth and enthusiasm*, agreement 1.0), consistent with the conservative-count caveat above.

| Feature | Judge-derived name                    | Agreement |
|---------|---------------------------------------|-----------|
| 148     | engaged and supportive                | 1.00      |
| 241     | formal, academic tone                 | 1.00      |
| 362     | empathetic and personally supportive  | 1.00      |
| 2036    | friendly, conversational warmth       | 1.00      |
| 89      | more empathetic and conversational    | 0.97      |
| 586     | enthusiastic and personally affirming | 0.97      |
| 717     | composed and confident                | 0.96      |
| 931     | formal and self-explanatory           | 0.96      |
| 709     | playful, humorous tone                | 0.95      |

Table 2: Persona directions the SAE discovered unsupervised and the judge named (above-null, 10 total; representative 9 shown). “Agreement” is the fraction of blinded judgements assigning the direction the same property.

**Reach.** We do not report prompt-elicitation numbers for the named directions here: as argued in §6, whether a discovered steerable direction can be reproduced by *some* prompt is close to unfalsifiable, and the useful quantity is the *graded* elicitation cost, which needs a calibrated protocol we leave to future work rather than a binary reach/no-reach label.

## 6 Discussion and limitations

The claim is scoped to one model, one layer, and one training mixture; the directions are causal under steering but the correspondence between an SAE feature and a human-named persona is judged, not proven. The screen’s consistency gate is sensitive to scenario count—a lesson from the pilot, and a parameter worth calibrating rather than fixing.

We deliberately do *not* claim to overcome the third requirement of Chen et al. [1], prompt-inducibility. A direction we discover and can steer is, almost by construction, reproducible by *some* prompt; “prompt-unreachable” is close to unfalsifiable and we avoid it. What the discovery result does establish is independence from the first two requirements—no trait is specified and no description is written—which is where an unsupervised dictionary has leverage a prompt-first method cannot. The sharper, and to us more interesting, question is the one Chen et al. [1] phrase themselves: whether SAEs surface directions “that cannot be *easily* elicited through prompting.” That word is graded, and it is where our reach probe points but does not yet reach: the right target is a calibrated *elicitation-cost* metric—how specific, how long, or how many attempts a prompt needs to match a discovered direction’s steering effect—that ranks directions from trivially prompt-inducible to stubbornly hard-to-elicite, and asks whether that hard tail is where unsupervised discovery earns its keep. That is the main thread of future work.

## 7 Conclusion

An unsupervised Matryoshka sparse autoencoder recovers persona-relevant directions in a chat model with no trait specified and no description written—dropping the first two requirements of prompt-based persona extraction—and gives a substrate for the sharper, graded question left open by the third: not whether a discovered direction can be prompted, but how easily. In a powered screen the method validated against 2 of 3 supervised controls and named 10 persona axes—warmth, empathy, formality, playfulness, confidence—that the model was never told to look for.

## References

- [1] R. Chen, A. Ardit, H. Sleight, et al. Persona Vectors: Monitoring and Controlling Character Traits in Language Models. *arXiv:2507.21509*, 2025.